VA DIVISION A • GRADES 3–4 • 2025–2026

Bungee Egg Drop

Competition Project Slides

- Official Rules
- Scoring Formula
- Artificial Horizon
- Bungee Design
- Energy Physics
- Competition Tips

Design a rubber band bungee system to stop a raw egg as close as possible to the Artificial Horizon without going past it.

Event at a Glance

2	25	2	50cm	4	100 pts
Students per team	Min build + test	Drops allowed	Tape provided	Band sizes	Max score

GOAL: Connect rubber bands into a bungee cord. Drop an egg (sealed in a plastic bag) from a two-story railing. Stop it as close to ZERO on the Artificial Horizon chart as possible — without going PAST zero.

Written test:	Taken SIMULTANEOUSLY during the 25-minute build phase. Topics: Conservation of Energy, GPE, KE, EPE.
Tape restriction:	Tape may ONLY attach bungee to bag. Cannot support or strengthen the egg or bag.
Two drops:	May adjust bungee between drops. Best drop score is used in scoring.
Safety:	ANSI Z87 D3 splash/droplet GOGGLES (full seal) required. NOT just impact glasses.

Materials:	Rubber bands (#19/#33/#64/#117B), 50cm masking tape, 1 large egg, 1 plastic bag.
Egg inspection:	Egg rolled on paper towel after drop. Any wet area = broken = Tier 2 (−20%).
Estimate:	Must announce estimated stop distance to supervisor. This is Tiebreaker 1!
Tier 3:	Did not follow Spirit of the Problem = −30% penalty.

Official Scoring Formula

Scoring

Final Score (FS) = Assessment Score (AS) + max(Drop Score 1, Drop Score 2) • Max = 100 points

Term	Formula	Notes
Assessment (AS)	50 × (your Raw Assessment Score ÷ best RAS among all teams)	Max 50 pts
Drop Score 1 (DS1)	50 × (TDD − DD1) ÷ max(TDD − DD for all teams)	Closer = higher
Drop Score 2 (DS2)	50 × (TDD − DD2) ÷ max(TDD − DD for all teams)	Best drop used
TDD (Target Drop Distance)	Announced distance from railing to Artificial Horizon. Minimum = 200 cm.	Set at event start
DD (Drop Distance)	200cm if egg passes below horizon; else distance from horizon to lowest egg point.	From slow-mo video
Tiebreaker 1	Smallest difference between your ESTIMATED and ACTUAL drop distance.	Announce carefully!
Tiebreaker 2	Correct answers on selected written assessment questions.	Know your formulas

★ *Example: TDD=400cm. RAS=6, best=9 → AS=33.33. 2nd drop DD=13cm, best DD=7cm → DS2=(400−13)÷(400−7)×50=49.24. Final Score = 82.57*

The Artificial Horizon — Explained

Horizon

ARTIFICIAL HORIZON CHART

+100cm	Too short — cord too stiff
+50cm	Good — close to target
0 = Horizon	PERFECT — maximum score
−50cm	Broke through — 200cm penalty
−100cm	Floor — DD = 200cm (worst)

The chart is mounted on a wall. Supervisor records the lowest point the egg reaches in slow-motion video.

1 Attach bungee to Retaining Hook

The END opposite the egg attaches to the hook. Rubber bands hang loose over front of platform.

2 Hold egg vertical at platform edge

Hold the egg (in bag) at the end of the jump platform in a VERTICAL position. Any holding style is fine.

3 Countdown and open hand

"3, 2, 1, drop" — OPEN hand. DO NOT push the egg. Any upward movement = violation.

4 Adjust between drops

Went past zero? Add shorter/stiffer bands. Stopped too high? Add longer bands or remove short ones.

5 Egg inspection

Remove egg from bag. Roll on paper towel. Any wet spot = broken = Tier 2. Cracks without leaking = OK.

Supervisor records: estimate, slow-mo video, distance to nearest 1cm, written assessment scores.

Bungee Design Strategy

Design

RUBBER BAND SIZES

- #19 — shortest, stiffest
- #33 — medium length/stiffness
- #64 — longer, softer
- #117B — longest, most stretch
- SERIES (end-to-end): longer cord, more total stretch
- PARALLEL (side-by-side): stiffer cord, less stretch per force
- Cut bands to make them shorter and stiffer

CALIBRATING YOUR BUNGEE

- At start: supervisor announces TDD (target distance)
- Calculate: bungee length + max stretch = approx TDD
- Test on floor: hang egg from fixed point, measure stretch
- Stopped short? Add longer bands or remove short ones
- Went past zero? Add shorter/stiffer bands
- First drop = data. Adjust. Second drop = score attempt.
- Write your estimate carefully — Tiebreaker 1!

ENERGY FORMULAS FOR THE TEST

- GPE = $m \times g \times h$ (gravity PE)
- KE = $\frac{1}{2} \times m \times v^2$ (kinetic energy)
- EPE = $\frac{1}{2} \times k \times x^2$ (elastic PE)
- At top: ALL energy is GPE
- Falling freely: GPE → KE
- Bands stretching: KE → EPE
- Lowest point: ALL energy is EPE

Competition Tips — How to Win

Tips

01

D3 goggles — not just glasses

Full-seal ANSI Z87 D3 (splash-rated) goggles are required. Regular impact glasses are NOT sufficient. Check "D3" on the label. Without them: no-show score.

02

Seal the egg completely

Seal the egg fully in the plastic bag before every drop. Any leakage after drop = broken = Tier 2 (−20%). Practice sealing quickly.

03

Know your estimate

Tiebreaker 1 is the closest estimate to actual drop distance. Time your test drops with a ruler and stopwatch. Write a precise number, not a guess.

04

Tape: attachment only

50cm of tape is ONLY for attaching the bungee to the bag. Cannot touch the egg or reinforce the bag. Violating this = Tier 2.

05

Use drop 1 as data

You get two drops. Don't try to win on drop 1 — observe carefully where the egg stops and adjust bands accordingly. Drop 2 should be your scored attempt.

06

Study the energy formulas

Assessment = 50 pts. Know GPE = mgh, KE = $\frac{1}{2}mv^2$, EPE = $\frac{1}{2}kx^2$. Know how they transform during the drop. The test happens while you build — review before you arrive.